Ascent of Rockot / Breeze-KM from Plesetsk

Flight data of two vehicle models on the 650 km reference mission

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Abstract. Two models of the Rockot launch vehicle with the Breeze-KM upper stage are flown from Plesetsk to a 650 km circular orbit at 86.6° with a three-degree-of-freedom ascent simulation. The first-principles model reaches a 648 × 676 km orbit after a maximum dynamic pressure of 66.5 kPa, and loses 1 110 m/s to gravity and 93 m/s to drag. Trajectory, atmospheric loads, propulsion, velocity losses and the orbit are reported for both models.

1 Introduction

This report documents a simulated ascent from Plesetsk (62.8° N, 40.3° E) to a 650 km circular orbit at 86.6°. The vehicle has three stages: Stage 1 (2 070.0 kN), Stage 2 (255.8 kN), Breeze-KM (19.6 kN), vacuum thrust, carrying a 1.57 t payload. The *first principles* model (solid lines in every figure) is built without reference to the User's Guide trajectory: engine performance from NASA CEA at the published chamber pressures and mixture ratios, propellant masses from engine mass flow and published burn times, drag from a component build-up, and steering optimised to leave the most Breeze-KM propellant in the target orbit. The *calibrated to the guide* model (dashed lines in every figure) fits the stage 1 and 2 propellant masses and the pitch programme to the reference trajectory and lift-off mass published in the Rockot User's Guide. Every number and figure below is generated from the simulation output by `aerosim.plots.ascent_report`; the same data drive the interactive dashboard.

2 Method

The vehicle is a point mass flown in an Earth-centred inertial frame over a rotating, oblate Earth,

$$\ddot{\mathbf{r}} = \mathbf{g}_{J_2}(\mathbf{r}) + \frac{F\,\hat{\mathbf{u}} - D\,\hat{\mathbf{v}}_{\mathrm{rel}}}{m}, \qquad \dot{m} = -\dot{m}_{\mathrm{stage}}, \tag{1}$$

with J_2 gravity, drag $D = \tfrac{1}{2}\rho v^2 C_D(M) A$ on the Earth-relative velocity in a co-rotating US Standard Atmosphere, and thrust corrected for nozzle back-pressure,

$$F = F_{\mathrm{vac}} - p_a A_e. \tag{2}$$

Mass flow is constant per stage and staging is instantaneous. The flight programme is a vertical rise, a pitch kick and a gravity turn for stage 1, a linear pitch programme for stage 2, and a closed-loop climb-rate law for the first Breeze-KM burn, which ends when the apogee reaches the target altitude; a second burn at apogee circularises the orbit. Velocity losses are integrated along the trajectory as

$$\Delta v_D = \int \frac{D}{m}\,\mathrm{d}t, \qquad \Delta v_p = \int \frac{p_a A_e}{m}\,\mathrm{d}t, \qquad \Delta v_g = \int g \sin\gamma\,\mathrm{d}t, \tag{3}$$

the gravity loss counted while the engines burn.

Table 1: Summary of the ascent. Losses are integrated over the powered flight to transfer-orbit insertion; the reserve is the Breeze-KM propellant left in the final orbit.

Quantity		First principles	Calibrated to the guide
Lift-off mass	t	110.63	116.24
Payload	t	1.57	1.57
Mach 1	s	33.1	33.2
Max. dynamic pressure	kPa	66.5	64.8
at time	s	52	51
at altitude	km	11.03	10.87
Peak acceleration	g_0	6.78	7.17
Insertion (transfer)	s	1 043.3	1 038.1
Ideal Δv, all burns	m/s	9 704	9 778
Gravity loss	m/s	1 110	1 152
Drag loss	m/s	93	70
Back-pressure loss	m/s	65	66
Perigee	km	648.2	642.9
Apogee	km	676.4	665.8
Inclination	deg	86.34	86.34
Period	min	97.98	97.82
Propellant reserve	kg	297	235

3 Results

3.1 Trajectory

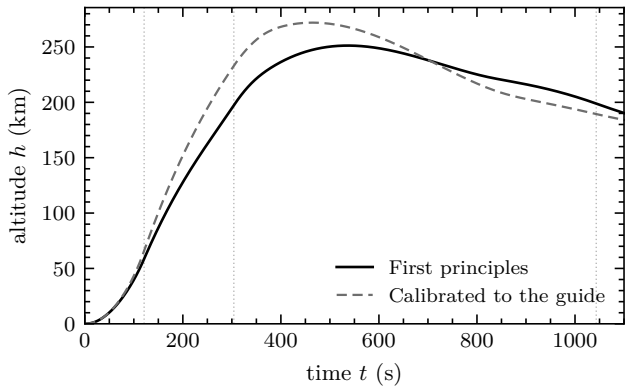
Table 1 collects the main figures of both flights. The vehicle passes Mach 1 at 33.1 s and maximum dynamic pressure, 66.5 kPa, at 52 s and 11.0 km (Figure 3). Acceleration peaks at 6.78 g_0 just before stage 1 burns out (Figure 6a). The fairing is released at 183.6 s and 115.2 km, once the free-molecular heat flux $\tfrac{1}{2}\rho v^3$ falls below 1 135 W/m^2 (Figure 4b). Breeze-KM cuts off at 1 043.3 s, 5 494 km down range, on a transfer orbit whose apogee sits at the target altitude; the second burn raises the perigee to 648 km, leaving an orbit of eccentricity 0.0020 at 86.34° (Table 2).

3.2 Atmospheric flight

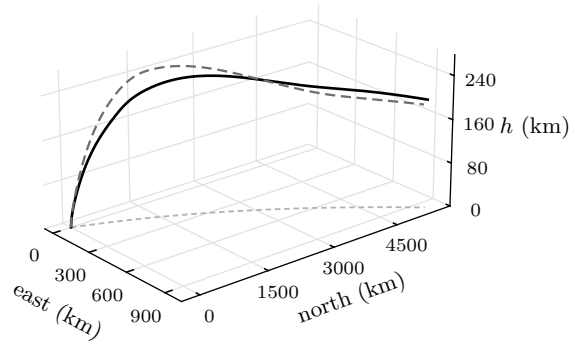
Drag and dynamic pressure are confined to the first 150 s (Figures 3 and 4). Drag climbs steeply through the transonic rise in C_D and peaks near maximum dynamic pressure.

3.3 Propulsion and losses

Ideal (vacuum rocket-equation) velocity summed over all burns is 9 704 m/s. Of it, 1 110 m/s is lost to gravity, 93 m/s to drag and 65 m/s to nozzle back-pressure (Figure 6b). The gravity loss falls back late in the

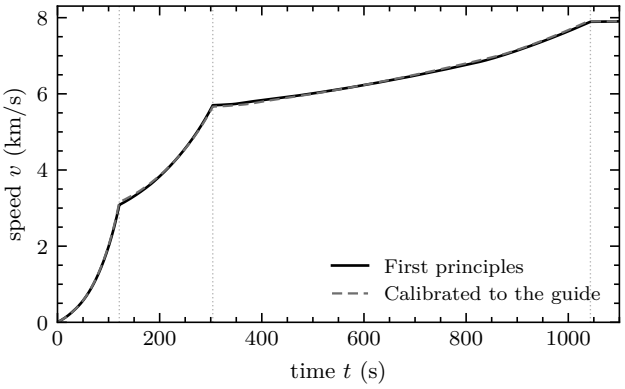


(a) Altitude to transfer-orbit insertion.

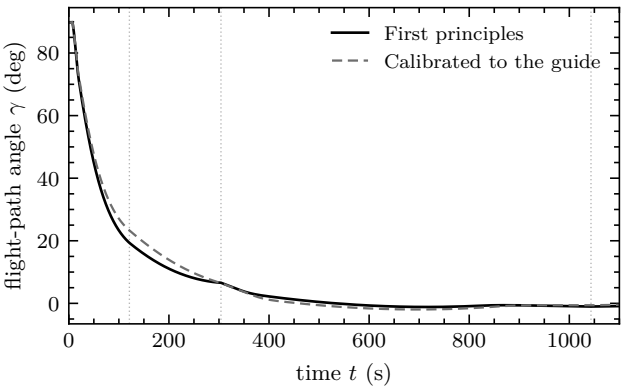


(b) Trajectory over the ground (azimuthal equidistant about the pad), with its projection.

Figure 1: Ascent from lift-off to Breeze-KM cut-off. Dotted verticals mark the stage cut-offs of the first model.

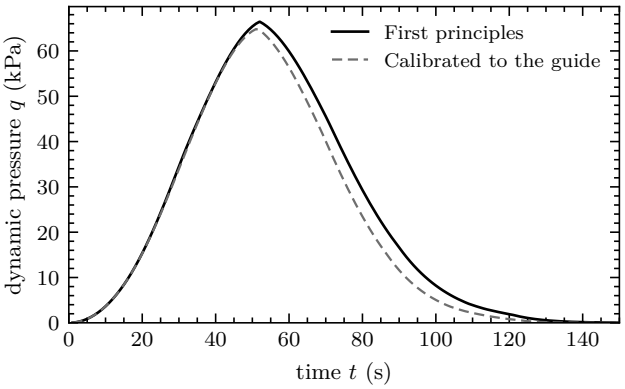


(a) Earth-relative speed.

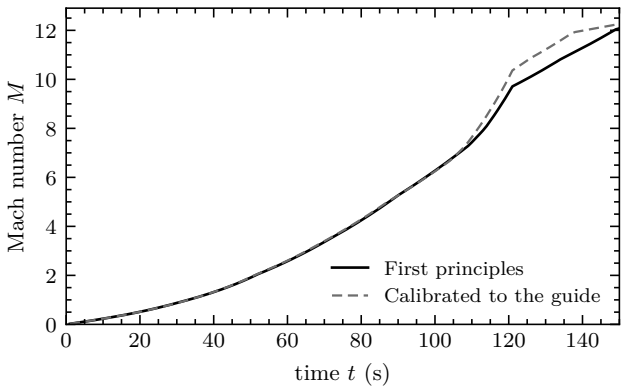


(b) Flight-path angle above the local horizon.

Figure 2: Velocity through the powered ascent.

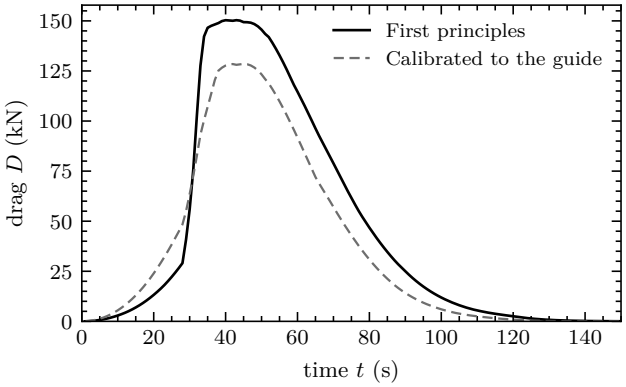


(a) Dynamic pressure.

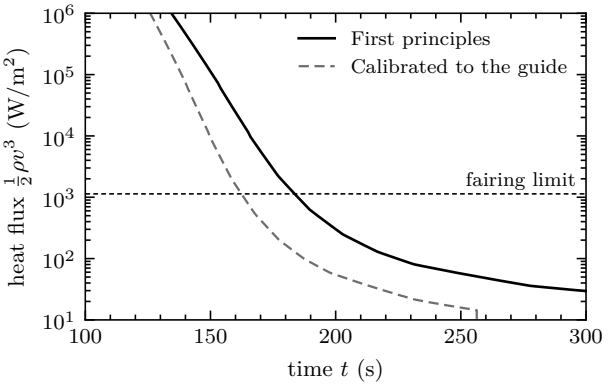


(b) Mach number.

Figure 3: The first stage through the dense atmosphere.



(a) Aerodynamic drag.



(b) Free-molecular heat flux after stage 1.

Figure 4: Aerodynamic loads. The fairing goes once the heat flux falls below the limit (dashed).

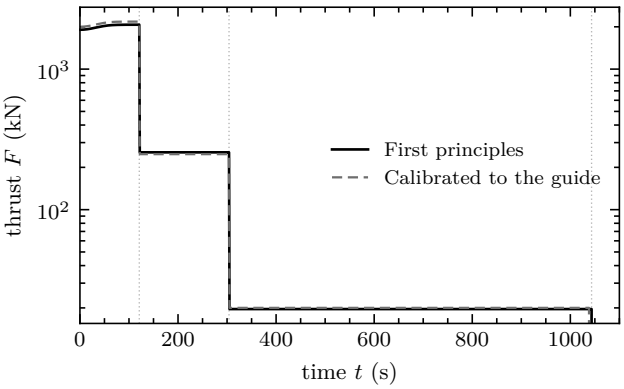
Breeze-KM burn, where the flight-path angle turns negative and gravity adds speed.

3.4 Orbit

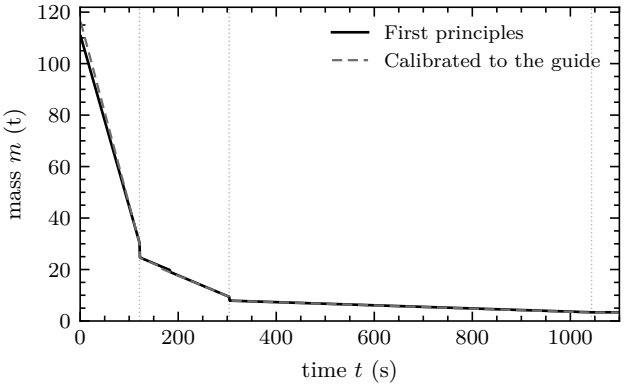
Tables 2 and 3 follow the orbit as it is built up stage by stage, and Figure 7 the ground track over the whole flight.

4 Discussion

Against the first principles model, the calibrated to the guide model lifts off 5.61 t heavier, peaks at 64.8 kPa of dynamic pressure (−2.5 %) and loses 1 152 m/s to gravity and 70 m/s to drag, against 1 110 and 93 m/s. It reaches a 643 × 666 km orbit with 235 kg of Breeze-KM propellant in reserve, against 648 × 676 km and 297 kg. Both flights rest on the same equations; the differences above come from the vehicle data and the flight programme each model uses.



(a) Thrust, logarithmic scale.



(b) Vehicle mass.

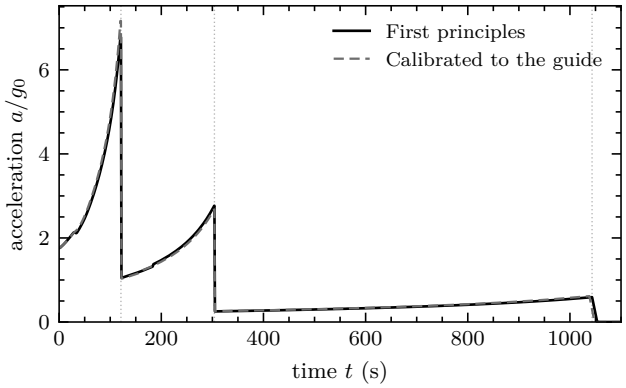
Figure 5: Thrust, with the back-pressure loss of each stage, and mass through staging and fairing jettison.

Table 2: Osculating orbit after each staging event: perigee and apogee altitude h_p, h_a (km), inclination i (deg) and eccentricity e. A negative perigee is a suborbital arc.

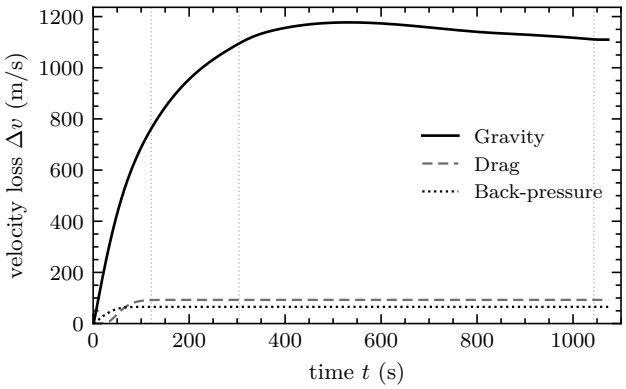
	First principles				Calibrated to the guide			
After	h_p	h_a	i	e	h_p	h_a	i	e
Stage 1 burnout and separation	−5 898	105	85.32	0.8621	−5 901	144	85.33	0.8637
Stage 2 burnout and separation	−4 004	228	86.31	0.4712	−4 014	262	86.31	0.4749
Transfer orbit	150	650	86.34	0.0369	160	635	86.34	0.0350
Final orbit	648	676	86.34	0.0020	643	666	86.34	0.0016

Table 3: Event timeline: time t (s), altitude h (km), Earth-relative speed v (km/s), flight-path angle γ (deg) and mass m (t).

	First principles					Calibrated to the guide				
Event	t	h	v	γ	m	t	h	v	γ	m
Lift-off	0.0	0.0	0.007	89.8	110.63	0.0	0.0	0.007	89.8	116.25
Max-Q	52.0	11.0	0.605	43.6	76.55	51.0	10.9	0.591	46.9	80.42
Stage 1 burnout and separation	121.0	58.6	3.083	19.3	30.44	121.0	65.9	3.156	23.3	30.27
Fairing jettison	183.6	115.2	3.639	12.4	19.04	162.4	114.2	3.464	18.0	20.60
Stage 2 burnout and separation	304.0	197.2	5.700	6.6	9.40	304.0	233.0	5.663	6.5	9.40
Breeze-KM ignition	304.0	197.2	5.700	6.6	9.40	304.0	233.0	5.663	6.5	9.40
Breeze-KM cut-off (transfer orbit)	1 043.3	198.8	7.892	−1.0	3.38	1 038.1	189.9	7.902	−0.6	3.31
Breeze-KM second burn	4 253.5	682.4	7.349	0.0	3.38	4 093.3	671.9	7.358	0.0	3.31
Breeze-KM cut-off (final orbit)	4 277.3	681.8	7.490	0.0	3.24	4 115.3	671.4	7.494	0.0	3.17

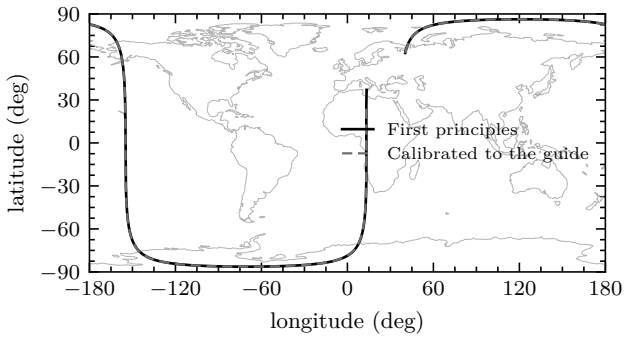


(a) Non-gravitational acceleration.

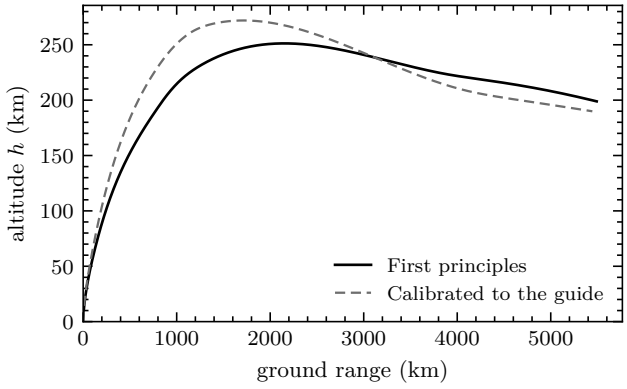


(b) Velocity losses, first principles model.

Figure 6: Axial load on the payload, and velocity lost to gravity, drag and back-pressure.



(a) Ground track over the full flight.



(b) Altitude against ground range.

Figure 7: Ground track (Natural Earth coastlines) and the ascent profile over the ground.